

PROJECT OBJECTIVE

Develop a multilingual NLP framework capable of:

1. Detecting coordinated influence operations.
 2. Detecting bot-like coordinated campaigns.
 3. Distinguishing:
 - Coordinated manipulation campaigns
 - Organic viral content
 - Satire content
 - Grassroots social movements
 4. Working across multiple languages.
 5. Supporting weakly-supervised and unsupervised learning.
 6. Producing explainable results.
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DATASETS

Use:

1. Twitter Information Operations Dataset (Stanford)
2. CREDBANK
3. MultiEURLEX

Create dataset loaders for each dataset.

Support:

- CSV
 - JSON
 - JSONL
 - TSV
-

TECH STACK

Python 3.11

Libraries:

- Pandas
- NumPy
- Scikit-Learn
- PyTorch
- HuggingFace Transformers
- Sentence Transformers
- NetworkX
- Plotly
- Streamlit
- UMAP
- HDBSCAN
- LangDetect
- FastText
- Optuna
- SHAP
- Seaborn
- Matplotlib

PROJECT ARCHITECTURE

Generate the entire project structure.

project_root/

|

| └─ app/

| | └─ streamlit_app.py

| | └─ dashboard.py

```
| ├── pages/
| | ├── 1_Data_Explorer.py
| | ├── 2_Preprocessing.py
| | ├── 3_Feature_Analysis.py
| | ├── 4_CIB_Detection.py
| | ├── 5_Network_Graph.py
| | ├── 6_Results.py
| | └── 7_Explainability.py
|
| ├── data/
| | ├── raw/
| | ├── processed/
| | ├── embeddings/
| | ├── networks/
| | └── samples/
|
| ├── src/
| | ├── data/
| | | ├── loader.py
| | | ├── merger.py
| | | └── validator.py
| |
| | ├── preprocessing/
| | | ├── language_detection.py
| | | ├── text_cleaning.py
| | | ├── translation.py
| | | ├── normalization.py
| | | └── preprocessing_pipeline.py
| |
| | ├── features/
| | | ├── textual_features.py
| | | ├── temporal_features.py
| | | ├── network_features.py
| | | ├── behavioral_features.py
| | | └── feature_fusion.py
| |
```

- | └─ models/
 - | | └─ embedding_model.py
 - | | └─ anomaly_detection.py
 - | | └─ graph_detection.py
 - | | └─ clustering.py
 - | | └─ weak_supervision.py
 - | | └─ cib_detector.py
- | |
- | └─ evaluation/
 - | | └─ metrics.py
 - | | └─ experiments.py
 - | | └─ multilingual_eval.py
- | |
- | └─ visualization/
 - | | └─ network_visualizer.py
 - | | └─ cluster_visualizer.py
 - | | └─ dashboard_charts.py
- | |
- | └─ utils/
 - | └─ config.py
 - | └─ logger.py
 - | └─ helpers.py
- |
- | └─ reports/
 - | └─ figures/
 - | └─ tables/
 - | └─ results/
 - |
- | └─ notebooks/
 - |
- | └─ tests/
 - |
- | └─ requirements.txt
- | └─ README.md
- └─ run.py

DATA PREPROCESSING

Implement:

1. Language Detection
2. Unicode normalization
3. Emoji handling
4. URL removal
5. Mention handling
6. Hashtag processing
7. Multilingual tokenization
8. Stopword removal
9. Lemmatization
10. Noise reduction

Support:

- English
 - French
 - German
 - Spanish
 - Arabic
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FEATURE ENGINEERING

Create four categories.

A. Textual Features

- Sentence embeddings
- Semantic similarity
- Topic modeling
- Sentiment

- Toxicity
- Linguistic diversity

B. Temporal Features

- Posting frequency
- Burstiness
- Synchronization
- Time-lag similarity

C. Network Features

- Retweet graph
- Mention graph
- Reply graph
- Community detection
- Centrality measures

D. Behavioral Features

- Account age
- Activity patterns
- Content repetition
- Hashtag repetition
- Link sharing behavior

CIB DETECTION MODEL

Build hybrid framework.

Stage 1

Sentence Transformer Embeddings

Use:

all-mpnet-base-v2

Stage 2

UMAP Dimensionality Reduction

Stage 3

HDBSCAN Clustering

Stage 4

Graph-based Coordination Detection

Use:

NetworkX

Detect:

- Dense communities
- Repeated content clusters
- Synchronized actors

Stage 5

Anomaly Detection

Use:

Isolation Forest

and

Local Outlier Factor

Stage 6

Weak Supervision

Generate pseudo labels using:

- Cluster confidence
- Graph density
- Temporal synchronization

EXPLAINABLE AI

Implement:

1. SHAP
 2. Feature Importance
 3. Coordination Score Explanation
 4. Cluster Explanation
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EVALUATION

Evaluate:

1. Precision
2. Recall
3. F1
4. Silhouette Score
5. Davies-Bouldin Score
6. Cluster Purity
7. Cross-Language Robustness

Generate publication-quality tables.

STREAMLIT DASHBOARD

Create professional UI.

Sidebar:

- Upload Dataset
- Select Language
- Run Pipeline
- Run Detection
- Run Evaluation

Pages:

1. Dataset Explorer
2. Preprocessing Results
3. Feature Analysis
4. Coordination Detection
5. Network Graph Visualization
6. Evaluation Results
7. Explainability

Include:

- Interactive Charts
 - Network Graphs
 - Cluster Maps
 - SHAP Visualizations
 - Download Reports
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SAMPLE DATA

Create sample datasets inside:

data/samples/

sample_twitter.csv

sample_credbank.csv

sample_multieurlex.csv

Populate each with realistic multilingual examples.

OUTPUT REQUIREMENTS

Generate:

1. Complete source code.
2. Every file separately.

3. Full documentation.
4. README.md.
5. requirements.txt.
6. Sample datasets.
7. Streamlit dashboard.
8. Installation guide.
9. Execution guide.
10. Research methodology documentation.

Provide code file-by-file with proper comments and professional software engineering practices.